

Summary of Personal, Relevant Background, and Future Goals Statement

Environmental hazards such as urban heat stress can be exacerbated by local pollution. The co-occurrence of such hazards is referred to as a compound event (1). However, such compound events have been sparsely studied. As a result, there is a knowledge gap in how compound environmental hazards such as urban heat stress and pollution can negatively impact public health. I want to advance our understanding of the nexus between street-level atmospheric chemistry processes and these regional environmental hazards through high-resolution urban environmental modeling. By doing so, I hope to support urban planning, policymaking, and health-based impacts assessments.

As such, over the next six years, I want to pursue a M.S. and a PhD in atmospheric sciences to **advance and develop this knowledge**. My research interests have been shaped by my educational training in meteorology, environmental sustainability, and sustainable energy systems. Professionally, I seek to become a tenured professor at a Carnegie R1 university, studying atmospheric chemistry processes in cities. Furthermore, I plan to become actively involved in the scientific policymaking landscape by pursuing educational opportunities that support my development as a science communicator. I strive for these goals because I want to advance knowledge, and **I want to mentor, inspire, and lead** these values and skills in my future research lab and with my own students. To date, my educational and supplementary involvement and successes have exceptionally prepared me to persevere through the challenges and journey that awaits as I pursue a PhD in atmospheric sciences.

As I seek to achieve these goals, I want to be under the mentorship of Dr. Chenghao Wang—PI of the Sustainable Urban Futures (SURF) Lab at the University of Oklahoma. The SURF Lab has pioneered cutting-edge numerical simulations of the urban environment that have enhanced our understanding of the interactions between urbanization and environmental hazards at various spatial and temporal scales. My success in this lab will be enhanced as I have already developed a close working relationship with the lab's PI and several of the lab's post-doctoral associates and current M.S. and PhD students. I have collaborated with the SURF Lab on several international and national conference presentations and manuscripts. These relationships will be crucial to help overcome the challenges I will face developing state-of-the-art high-resolution urban environmental models. I will inevitably need a strong network of support, and importantly, a mentor who understands my career goals and who will help me achieve them.

I have lead-authored and published two manuscripts, **secured an undergraduate research grant** to study the relationship between convective weather and ozone pollution in Oklahoma City, Oklahoma, and have held research internships at the nation's premier agencies responsible for studying and managing atmospheric chemistry: the National Oceanic and Atmospheric Administration (NOAA) Chemical Sciences Laboratory (CSL) and the Environmental Protection Agency Radiation Protection Division. I have demonstrated exceptional commitment and achievement to my research and professional goals that are **a direct result of the rigorous scientific challenges and experiences that I have faced and that are furthermore extraordinarily uncommon for someone at my career stage**. In 2025, I received the Barry Goldwater Scholarship for my research in evaluating an advanced, high-resolution, convection-permitting numerical weather prediction model. This project saw hours of testing hypotheses, debugging code, collaboration, and seeking insight from scientists at other institutions. It was a process that **demonstrated my perseverance and capabilities to solve complex challenges**. This prestigious award is made to less than 10% of applicants and is a result of my engagement with the rigorous scientific process.

In further support of my ambition to pursue a PhD and **have a high-societal impact**, I earned a premier recognition from NOAA, recognizing my outstanding scientific and academic achievement, excellence, and future potential to advance NOAA mission goals. This recognition was earned through the NOAA Hollings Scholarship, a competitive, merit-based honor awarded to less than 30% of applicants. The Hollings Scholarship recognizes my ability to **set a goal, achieve it, and support** the agency's scientific mission and philosophy.

Through the Hollings program, I completed an intensive nine-week internship in Boulder, Colorado, at NOAA CSL. Here, I developed and integrated a meteorological evaluation toolkit and paired it with a chemistry model evaluation toolkit. Atmospheric scientists and chemists alike at CSL expressed

their needs for the ability to simultaneously evaluate the chemistry and atmospheric simulations of their models. I set a goal to develop this meteorological evaluation toolkit, and as a result, ***I led the development and implementation of a solution*** to their scientific need.

I used CSL's existing chemistry model evaluation framework and developed a model evaluation toolkit that contained the ability to conduct atmospheric analyses that would be useful to chemists. For example, while atmospheric scientists and atmospheric chemists are concerned with temperature, a more useful representation of temperature to atmospheric chemists is potential temperature. However, there was not a consistent calculation of surface and vertical potential temperature across CSL's atmospheric models and observations. My toolkit provided atmospheric chemists at CSL a method to consistently calculate such variables across all their models and observations—vertically, horizontally, and agnostic to the model and observation type. I identified a scientific need, and I met that need with a solution. This will be a hallmark throughout my career.

Not only do ***I have the perseverance to face challenges and innovate solutions*** to them, but I know how to share the knowledge and original contributions I make within this field. I lead-authored a publication in *Urban Climate* where I highlighted several key weaknesses of using simplified urban canopy models to represent the complex structure of the built urban environment, especially when needing to simulate variables that influence human comfort and energy demand (2). Further, I lead-authored a literature review on the communication of uncertainty in Earth system models (ESMs) published in *Weather Climate and Society* (3). I saw a need to provide atmospheric scientists a succinct subject review of how uncertainty in ESMs is communicated, and I saw a lack of representation for how decision-makers perceive and believe uncertainty to be communicated by atmospheric scientists. As such, I believed I would have an outsized impact by lead-authoring such a review for atmospheric scientists and decision-makers alike. These are key skills that give me a unique advantage in ensuring my success and professional growth throughout graduate studies.

My first-hand experience in the scientific publication process will be invaluable as I will lead-author several more papers throughout my M.S. and PhD. Further, I have extensive experience with and have ***won outstanding presentation awards*** for the scientific presentation of my work at international, national, and regional conferences. Importantly, these skills and experiences will enable a more efficient working relationship with my mentor as we would ***develop a mentoring plan*** that can refine and sharpen the skills I need as a prospective professor.

I have also demonstrated an exceptional commitment to my community and nation through impacts-based service, further reinforcing ***my potential to benefit society and advance knowledge***. My passion and commitment to atmospheric science have gone so far that it led me to Washington D.C. to advocate for U.S. Senate sponsorship of the Weather Authorization Reauthorization Act (H.R. 5089) as part of the American Geophysical Union's Voices for Science Fellowship. This gave me an invaluable opportunity to experience scientific policymaking firsthand and was the result of an exceptionally competitive global pool of applicants. Through this program, I learned that I want to actively work with legislators on scientific policy—not only in my career, but as a graduate student.

Because speaking with policymakers is an experience I have, I will use my individual mentoring plan to hone and further develop my skills as a confident science communicator. Additionally, I would apply these skills by demonstrating the real-world policymaking and decision-making implications of my graduate research on compound environmental hazards. This level of outreach will be geared towards a broad audience, ranging from the public to the policymakers themselves.

Intellectual Merit

I am an ***emerging, early-career interdisciplinary atmospheric chemist*** with focuses on understanding the urban environment. In addition to my pre-existing network at the SURF Lab, I have built strong collaborations with scientists at NOAA CSL and the National Center for Atmospheric Research Applications Laboratory. I will be able to leverage these networks and collaborations to support their scientific needs outside the SURF Lab, and I will be able to receive additional mentorship and

guidance to the challenges I will face. The 2025 Barry Goldwater Scholar and 2024 NOAA Hollings Scholar awards already place me among an exceptionally small group of emerging, early-career atmospheric scientists. As a dual major in meteorology and environmental sustainability and as an individual trained in atmospheric science who can ***critically and creatively solve the challenges*** faced by atmospheric chemists, that group becomes exceptionally smaller. Additionally, I know with fervor and passion that I want to contribute to the mentorship and upbringing of globally-competitive atmospheric scientists in my own developed research lab.

With my background experience to date, ***I have taken initiative and developed clear, specific goals*** that will support my professional ambitions of becoming an atmospheric chemist and a professor. Additionally, ***I have developed explicit training goals*** and professional development experiences that I want to have incorporated into an individual mentoring plan as a future graduate student. The evolution of my scientific skills outside the lab is crucial for my professional development to prepare me to advance knowledge and mentor the next generation of students across the atmospheric sciences. My scientific background, experience, and goals as an emerging, early-career scientist are nothing short of a demonstration of my intellectual merit.

Broader Impacts

I want to use my educational training to become a professor, which will enable me to ***support and mentor the next generation of atmospheric scientists***. I would have a unique impact on the educational upbringing of students that enroll in future courses I teach. I would also have an exceptional impact on being able to shape future students' understanding of the interdisciplinary importance of atmospheric science and chemistry to shared interests of protecting life, property, and ***enhancing economic prosperity***.

My research interests in developing high-resolution urban environmental models would support health-based, city-level impact assessments of compound environmental hazards. Critically, my work would enable smarter, more efficient urban planning and implementation of solutions to reduce environmental pollution. From these results, decision-makers from individual municipalities to entire nations could implement real-world mitigation strategies that will improve the well-being of their cities. I would be conducting this work in pursuit of advancing such national interests, but I would also have developed the skills to directly work with policymakers who can command U.S. science policy. Working directly with the policymakers is the type of work that ensures continuous scientific advancement, which as a result will ***improve the well-being of society***.

My research interests and professional goals have immeasurable potential to have broader impacts for the better of society. I am dedicated to teaching, mentoring, and the pursuit of scientific excellence. Combined, these are a rare combination of traits to possess as a scientist, which is an extraordinary opportunity for me to change the long-standing paradigm that scientists are inaccessible. My commitment to public service and ensuring that my work has broad societal impacts are underscored by having earned the 2025 Barry Goldwater and 2024 NOAA Hollings awards. As an emerging, early-career scientist, I am ready and have been rigorously ***prepared to confront the nation's largest scientific challenges*** across atmospheric science and chemistry.

References

[1] C. Wang et al., *Urban Clim.* 62, 102511 (2025). [2] L. Thompson et al., *Urban Clim.* 61, 102375 (2025a). [3] L. Thompson et al., *Weather Clim. Soc.* (2025b).